

STROKE

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Case

- **CC:** “My right arm feels like it’s frozen. I can barely move it.”
- **HPI:** Mr. B is a **67-year-old Ethiopian man** who presents to the emergency room at 8:45 AM after noticing a **sudden onset of weakness in his right arm**. He woke up at 7:15 AM and went to the bathroom to brush his teeth. While walking from the bathroom to the kitchen, he noticed **general weakness and had trouble saying “good morning” to his son**, with whom he lives. His son immediately brought him to the ER. While in the ER, he started experiencing some dysarthria and began to have a **right sided facial droop**. He denied any dizziness, vomiting, or **headache**.

What do you think the health problem of Mr. B?

Stroke

- ***Stroke** is a term used to describe an abrupt onset of **focal neurologic deficit** that lasts at least 24 hrs and is presumed to be of **vascular origin**.*
- Stroke “cerebrovascular accident (CVA) ” or “brain attack”
- Transient ischemic attacks (**TIAs**) are focal ischemic neurologic deficits lasting less than 24 hrs and usually less than 30 min.

Epidemiology

- **Stroke** is the 3rd leading killer worldwide and the 5th leading cause of death in the Ethiopia,
- Stroke risk is increased above that of the general population in **elderly male** individuals and in AfAm.
- Stroke is the 3rd leading cause of **disability**.

WHO (2021), Global

WHO (2021), Ethiopia

Top 10 global causes of death in 2021

1. Ischaemic heart disease
2. COVID-19
3. Stroke
4. Chronic obstructive pulmonary disease
5. Lower respiratory infections
6. Trachea, bronchus, lung cancers
7. Alzheimer disease and other dementias
8. Diabetes mellitus
9. Kidney diseases
10. Tuberculosis

Top 10 global causes of disability-adjusted life years (DALYs) in 2021

1. COVID-19
2. Ischaemic heart disease
3. Stroke
4. Lower respiratory infections
5. Preterm birth complications
6. Back and neck pain
7. Diabetes mellitus
8. Chronic obstructive pulmonary disease
9. Diarrhoeal diseases
10. Road injury

Top causes of death

Deaths per 100 000 population, Ethiopia, 2021

Lower respiratory infections	47.7	
Preterm birth complications	34.5	
Diarrhoeal diseases	33.7	
COVID-19	31.9	
Stroke	29.7	
Birth asphyxia and birth trauma	25	
Ischaemic heart disease	20.5	
Road injury	17.7	
Tuberculosis	17.3	
Cirrhosis of the liver	16.2	

Prevalence of Stroke in Ethiopia

- **27** studies in this systematic review and meta-analysis.
- The pooled burden of hemorrhagic and ischemic stroke were **46.42%** (95%CI: 41.82–51.53; $I^2 = 91.6\%$) and **51.40%** (95%CI: 46.97–55.82; $I^2 = 85.5\%$) respectively.
- The overall magnitude of modifiable risk factor:
 - Hypertension (49%), alcohol consumption (24.96%) and dyslipidemia (20.99%) among stroke patients

Weldegiorgis TA, et al.,(2021). *The burden of stroke and modifiable risk factors in Ethiopia: A systemic review and meta-analysis.* **PloS one, 16(11), e0259244.**

RISK FACTORS FOR STROKE

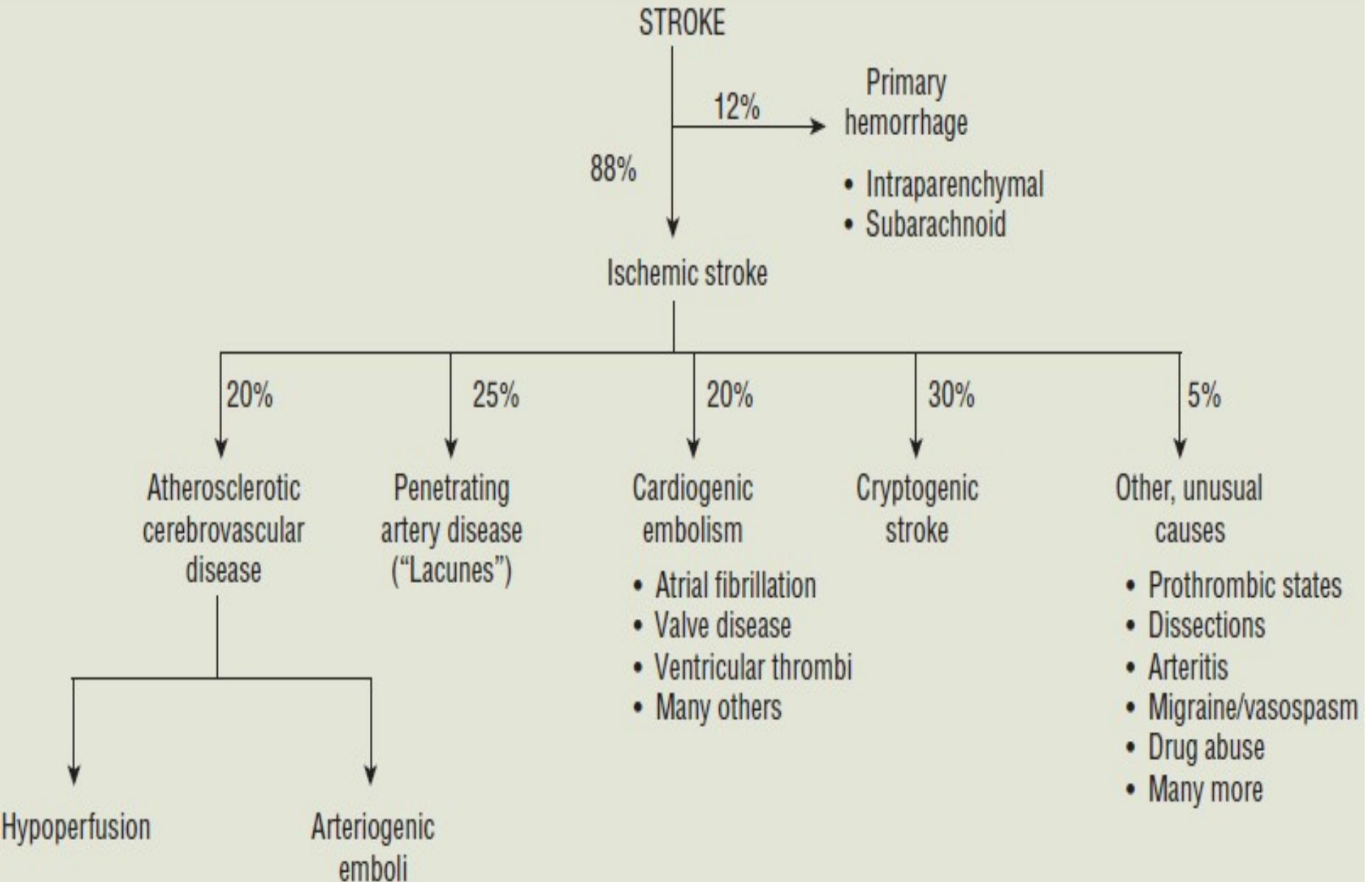
Nonmodifiable risk factors

- Increased age (>55yr)
- Male gender (males)
- Race (AfAm, Asian Hispanic)
- Family history (hereditary)
- Low birth weight

Modifiable risk factors

- **Hypertension**
- **Cardiac disease** (esp. **AF**)
- **Diabetes mellitus**
- Dyslipidemia
- Cigarette smoking
- Alcohol
- Life style (obesity, physical inactivity, diet)

Classification of Stroke



- Ischemic strokes (87%) are due either to **local thrombus** formation or to **emboli** that occlude a cerebral artery.
- These result **arterial occlusion**, decreasing cerebral blood flow and causing ischemia and ultimately infarction distal to the occlusion
- Atherosclerosis of large intracranial or extracranial arteries or small artery disease can result in ischemic stroke.
- Emboli can arise from the heart in patients with **atrial fibrillation**, **valvular heart disease**, or other **prothrombotic heart problems** and cause about 25% of ischemic strokes.
- The stroke cause is undetermined in some cases

- Proposed mechanisms of occlusion that lead to ischemic stroke
 - ✓ **Atherosclerosis**
 - ✓ Penetrating artery disease
 - ✓ Cardiogenic emboli
 - ✓ Cryptogenic stroke
 - ✓ Other causes

- Hemorrhagic strokes account for 13% of strokes and include:
 - Subarachnoid hemorrhage (SAH),
 - Intracerebral hemorrhage (ICH)
- **SAH** may result from **trauma or rupture** of an intracranial aneurysm or arteriovenous Malformation (AVM).
- ICH occurs when **bleeding** in the brain parenchyma results in hematoma formation.
- The presence of blood in the brain causes damage to surrounding tissue through
 - **Mass effect** and **Neurotoxicity of blood components** and their degradation products.

Transient Ischemic Attack (TIA)

- “Mini-stroke”
- Describes a temporary focal neurologic deficit lasting < 24 hrs (typically 2 to 5 minutes)
- Caused by **brief interruption of blood supply** to part of the brain
- 35% of untreated patients will develop a stroke within 5 years of TIA

- The pt may experience:
 - Sudden weakness on one side of the body,
 - Inability to speak (**slurred speech**),
 - Loss of vision/**blurred vision**,
 - Vertigo, or falling
 - Facial drop
 - Difficulty swallowing
- Ischemic stroke is usually painless, but headache may be severe in hemorrhagic stroke

Facial Droop



Normal



Abnormal

Diagnosis

- Pt history from family or witness
- **Computed Tomography (CT) Scan:** Reveal areas of hemorrhage (**white**), Ischemia may not be visible up to 24 hr
- **Magnetic Resonance Imaging (MRI):** Reveal areas of ischemia with higher resolution and earlier than the CT scan
- **CT and MRI** head scans can reveal areas of hemorrhage and infarction
- Computed tomography angiography (CTA) is recommended in patients with endovascular treatment indications
- **Carotid Doppler studies:** Determine degree of stenosis in the carotid arteries
- **ECG:** Determine presence atrial fibrillation
- **Transthoracic echocardiogram (TTE),** and transcranial Doppler (TCD) studies can each provide valuable diagnostic information.

- **Blood glucose**, platelet count, and coagulation parameters (eg, prothrombin time, aPTT) are used in stroke assessment to determine treatment eligibility.
- Tests for **hypercoagulable states (protein C and S deficiency**, antiphospholipid antibody) should be done only when the etiology cannot be determined based on presence of well-known risk factors.

Treatment

- **Goal of therapy:**
 - (1) Reduce the ongoing neurologic injury & decrease mortality and long-term disability
 - (2) Prevent complications 2^o to immobility and neurologic dysfunction; and
 - (3) Prevent stroke recurrence.

- The initial approach is to ensure
 - Adequate respiratory and cardiac support
 - To determine **quickly** whether the lesion is ischemic or hemorrhagic based on a CT scan.
- Ischemic stroke pts presenting within **hrs** of symptom onset should be evaluated for **reperfusion therapy**
- * **Elevated BP should remain untreated** in the acute period (first 7 days) after ischemic stroke because of the risk of decreasing cerebral blood flow and worsening symptoms.

- Patients with TIA require urgent assessment and intervention to reduce the risk of stroke, which is highest in the first few days after TIA.
- Pts with hemorrhagic stroke should be assessed to determine whether they are candidates for **surgical intervention**.
- After the acute phase, attention is focused on:
 - Preventing progressive deficits,
 - Minimizing complications, and
 - Instituting appropriate secondary prevention strategies.

Ischemic Stroke

- Endovascular intervention and thrombectomy with retrievable stents to **reperfuse ischemic brain tissue--**
- Thrombectomy is strongly recommended- within **6 hrs** of symptom onset (6-24 hr in selected patients)
- **Decompressive hemicraniectomy** is a surgical procedure to reduce intracranial pressure (typically due to cerebral edema) and can reduce mortality and improve functional outcome in select patients.
- **Early rehabilitation (MD)** is very effective in reducing long-term disability.
- In **secondary prevention**, **carotid endarterectomy** of an ulcerated or stenotic carotid artery is effective in reducing stroke incidence and recurrence in appropriate patients
- In patients **younger than age 70**, **carotid stenting** is a **less invasive** alternative and can reduce recurrent stroke risk when combined with **aspirin and clopidogrel** therapy.

Hemorrhagic Stroke

- In **SAH** from ruptured intracranial aneurysm or AVM, early intervention with either **surgical clipping** or **endovascular coiling** of the vascular abnormality reduces mortality from rebleeding.
- **Early surgical intervention and hematoma removal** is recommended for patients with cerebellar hemorrhage and neurologic deterioration, brainstem compression, or hydrocephalus from ventricular obstruction. The usefulness of surgical hematoma evacuation is not well established for patients with **cerebral hemorrhage**.
- Ventricular drainage with an extraventricular drain (**EVD**) is reasonable in patients with hydrocephalus causing decreased consciousness

Pharmacologic Therapy Stroke

Acute Ischemic Stroke

- **Alteplase** (IV tissue plasminogen activator)
 - Initiated within **4.5 hrs** of symptom onset has been shown to reduce the ultimate disability due to ischemic stroke.
 - The dose is **0.9 mg/kg** (maximum 90 mg) infused IV over **1 hr** after bolus of 10% of the total dose given over 1 min
- **Aspirin 160 to 325 mg/day** started between 24 - 48 hrs of symptom onset or (24 hrs after completion of alteplase) has also been shown to reduce long-term death and disability.

Recommendations for Pharmacotherapy of Ischemic Stroke

	Recommendation	Evidence ^a
Acute treatment	Alteplase 0.9 mg/kg IV (maximum 90 mg), 10% as a bolus with remainder given over 1 hour in select patients within 3 hours of onset	IA
	Alteplase 0.9 mg/kg IV (maximum 90 mg), 10% as a bolus with remainder given over 1 hour in select patients between 3 and 4.5 hours of onset	IB
	Aspirin 160–325 mg daily started within 48 hours of onset	IA
Secondary prevention		
Noncardioembolic	Antiplatelet therapy	
	Aspirin 50–325 mg daily	IA
	Aspirin 25 mg + extended-release dipyridamole 200 mg twice daily	IB
	Clopidogrel 75 mg daily	IIaB
Cardioembolic (especially atrial fibrillation)	Anticoagulant therapy	
	Vitamin K antagonist (warfarin; target INR = 2.5)	IA
	Apixaban 5 mg twice daily	IA
	Dabigatran 150 mg twice daily	IB
	Edoxaban 60 mg daily	— ^b
	Rivaroxaban 20 mg daily	IIaB
Age ≤75 years	High-intensity statin therapy	IB
Age >75 years	Moderate- or high-intensity statin therapy	IIaB-R
BP >140/90 mm Hg	BP reduction	IB-R

Inclusion criteria

- Age 18 years or older
- Clinical diagnosis of ischemic stroke with neurologic deficit
- Time of symptom onset well established to be <4.5 hours before treatment would begin

Contraindications

- Symptoms/imaging consistent with SAH or acute intracerebral hemorrhage
- Current use of direct thrombin inhibitors or direct factor Xa inhibitors in prior 48 hours
- Use of treatment-dose low-molecular-weight heparin in prior 24 hours
- Infective endocarditis
- Intra-axial, intracranial neoplasm
- Aortic arch dissection
- Active internal bleeding or coagulopathy (platelets <100,000/mm³ [100 x 10⁹/L], INR >1.7, aPTT >40 sec, PT >15 sec)
- Severe head trauma in prior 3 months
- Gastrointestinal malignancy or bleeding within prior 21 days

Warnings/Use Clinical Judgment

- History of intracranial hemorrhage
- History of ischemic stroke within prior 3 months
- Unruptured/unsecured AVM or aneurysm >10 mm
- Major surgery or nonhead trauma
- History of bleeding diathesis
- Extensive regions of clear hypoattenuation on initial CT scan

- For pts with **elevated BP who are eligible for alteplase**, treatment to a goal BP <185/110 mm Hg is recommended before thrombolytic administration. If BP is treated, short-acting and easily titrated IV agents are preferred:
 - ✓ **Labetalol**: 10–20 mg IV over 1–2 minutes; may repeat
 - ✓ **Nicardipine**: 5 mg/hr IV; titrate up by 2.5 mg/hr every 5–15 minutes; maximum 15 mg/hr
 - ✓ **Clevidipine**: 1–2 mg/hr IV; titrate by doubling the dose every 2–5 minutes; maximum 21 mg/hr
 - ✓ Other potential agents: **hydralazine, enalaprilat, nitroprusside IV infusion, labetalol IV infusion**

- Use of **anticoagulation** (eg, UFH or LMWH) is **not routinely** recommended in acute ischemic stroke treatment
- Use of immediate anticoagulation for non-stroke indications (eg, prophylaxis of DVT should be weighed against the risk of intracranial hemorrhagic conversion).

- All patients who have had an acute ischemic stroke or TIA should receive **long-term antithrombotic therapy** for secondary prevention.
- **Antiplatelet therapy** is cornerstone for Secondary prevention of Noncardioembolic ischemic stroke as
- **First-line antiplatelet agents**
 - **Aspirin** 50-325 mg daily, or **Clopidogrel** 75mg daily, and
 - **Extended-release dipyridamole** (200mg) + **aspirin** (25 mg) twice daily.
- For patients with **atrial fibrillation** and a presumed cardiac source of embolism for stroke or TIA, **oral anticoagulation with a vitamin K antagonist (warfarin, INR = 2.5)**, apixaban, dabigatran, edoxaban, or rivaroxaban is recommended

BP Control

- Adults previously HTN treatment, start after the first few days of the index event to reduce the risk of recurrent stroke and other vascular events.
- Useful options include a **thiazide diuretic, ACE Inhibitors, angiotensin receptor blocker**, or **combination treatment** with a thiazide Plus ACE inhibitor.
- Selection of specific drugs should be individualized based on patient comorbidities.
- Adults not on HTN who experience a stroke or TIA and have a BP $\geq$ 140/90 mm Hg should be prescribed antihypertensive treatment several days after the index event.
- A reasonable goal BP is **$\leq$ 130/80** mm Hg

Statin therapy

- **Statin therapy is recommended to prevent stroke recurrence in all ischemic stroke** patients regardless of baseline lipid levels.
- Patients ≤ 75 years of age experiencing ischemic stroke of presumed atherosclerotic origin should be treated with **high-intensity statin therapy** with a target of achieving $\geq 50\%$ reduction in low-density lipoprotein (LDL) cholesterol.
- For patients older than age 75, moderate- or high-intensity statin therapy can be initiated as tolerated.
- **Ezetimibe may be added for patients** taking maximally tolerated statin therapy but with LDL cholesterol ≥ 70 mg/dL (1.81 mmol/L).

- The NCEP* considers ischemic stroke or TIA to be a coronary risk equivalent and recommends the use of **statins** in ischemic stroke pts to achieve LDL-C < 100 mg/dL.
- **LMWH** or **low-dose SC unfractionated heparin** (7,500 U BID) is recommended for prevention of **deep venous thrombosis** in hospitalized pts

NCEP*- National cholesterol education program

PHARMACOLOGIC THERAPY OF HEMORRHAGIC STROKE

- There are currently **no standard pharmacologic strategies** for treating spontaneous ICH.
- Managing BP, increased ICP, and other medical complications in acutely ill pts in neurointensive care units should be followed.
- The **CCB, nimodipine** is recommended to reduce the incidence and severity of neurologic deficits resulting from delayed ischemia from subarachnoid hemorrhage
- Because HTN increases the risk of hematoma expansion, pts with SBP>220 mmHg to receive aggressive BP lowering with continuous IV infusion
 - BP goal: <140mmHg,
- SAH due to aneurysm rupture, BP control to at least a SBP<160 mm Hg
- When intracranial hemorrhage occurs in a pt on anticoagulants, consider reversal agents to correct the medication-induced coagulopathy.

- Monitored intensely for **neurologic worsening** (recurrence or extension of stroke), **complications (VTE, Infection)**, and adverse effects from treatments
- The most common reasons for clinical deterioration in stroke pts are:
 - (1) Extension of the original lesion in the brain;
 - (2) Development of cerebral edema and raised ICP
 - (3) Hypertensive emergency
 - (4) Infection (urinary and respiratory tract infections)
 - (5) Venous thromboembolism
 - (6) Electrolyte abnormalities and rhythm disturbances
 - (7) Recurrent stroke